

Complete Scientific Data Foundation & Official Institutional Provenance Guide

This document provides the definitive, point-by-point scientific foundation for all parameters, distributions, statutory formulas, and empirical moments utilized in this research platform.

Pillar 1: Statutory Pension Formula & 2026 Actuarial Constants (SGB VI)

- **Official Institution:** Federal Ministry of Labour and Social Affairs (*Bundesministerium für Arbeit und Soziales - BMAS*) & *Deutsche Rentenversicherung Bund (DRV)*.
- **Legal Codes & Ordinances:**
 - *Sozialgesetzbuch VI (SGB VI)*: Statutory Pension Insurance.
 - *Rentenwertbestimmungsverordnung 2024-2026 (Bundesgesetzblatt / BGBl.)*.
 - *Sozialversicherungs-Rechengrößenverordnung 2026*.
- **Exact Calibrated Parameters:**
 - **Aktueller Rentenwert ($AR_{\{2026\}}$):** EUR 42.52 per Entgeltpunkt (post-July 2026 unified national value, § 68 SGB VI).
 - **Durchschnittsentgelt ($DE_{\{2026\}}$):** EUR 51,944 preliminary reference wage for 1.0 EP (§ 68 / Anlage 1 SGB VI).
 - **Beitragsbemessungsgrenze ($BBG_{\{2026\}}$):** EUR 101,400/year (EUR 8,450/month), setting the statutory contribution ceiling at $EP_{\max} = 1.9521$ (§ 159 SGB VI).
 - **Contribution Rate (*Beitragssatz*):** 18.60% split equally between employer and employee (9.30% each, § 158 SGB VI).
 - **Childcare Credits (*Kindererziehungszeiten*):** Up to 3.0 EP (~EUR 127.56/month) per child born 1992+ credited to the primary caregiver (§ 56 SGB VI).
 - **Basic Pension Supplement (*Grundrente*):** Up to 0.8 EP/year uplift (max 33.4 EP cap) for workers with 33+ qualifying years earning 0.3-0.8 EP/yr (§ 76g SGB VI).
 - **Standard Pensioner Benchmark (45 EP):** Gross EUR 1,913.40/month *↪* Net EUR 1,677.10/month (parent) / EUR 1,665.62/month (childless).
 - **Mandatory Retiree Deductions:**
 - Health Insurance for Pensioners (*KVdR*): 8.75% (7.30% statutory base + 50% of average Zusatzbeitrag 2.90% = 1.45%, SGB V § 241/247).
 - Long-Term Care Insurance (*PV*): 3.60% for parents / 4.20% for childless pensioners (SGB XI § 55).
- **Exact Official URLs:**
 - [SGB VI Full Statutory Text \(Gesetze im Internet\)](#)
 - [DRV Official Pension Values & Parameters](#)
 - [BMAS Pension Ordinance Portal](#)

Pillar 2: Social Assistance & Subsistence Floor (SGB XII, Chapter 4)

- **Official Institution:** Federal Ministry of Labour and Social Affairs (*BMAS*).
- **Legal Codes & Ordinances:**
 - *Sozialgesetzbuch XII (SGB XII)*: Social Assistance (*Sozialhilfe*), Chapter 4 (*Grundsicherung im Alter und bei Erwerbsminderung*).
 - *Regelbedarfsstufen-Fortschreibungsverordnung*.
- **Exact Calibrated Parameters:**

- **Standard Subsistence Rate (*Regelbedarfsstufe 1*):** EUR 563/month for a single adult (§ 27a / § 28 SGB XII).
- **Accommodation & Heating Costs (*Kosten der Unterkunft und Heizung - KdU*):** Modeled at the federal average single benchmark of EUR 550/month (§ 35 / § 42 SGB XII).
- **Total Illustrative Single Subsistence Benchmark:** EUR 1,113/month (EUR 563 standard rate + EUR 550 KdU benchmark).
- **Protected Liquid Assets (*Schonvermögen*):** EUR 10,000 exemption for a single adult (§ 90 Abs. 2 Nr. 9 SGB XII).
- **Grundrente Income Disregard (*Freibetrag nach § 82a SGB XII*):** Up to EUR 281.50/month (50% of Regelbedarfsstufe 1) excluded from means-testing for individuals with 33+ Grundrentenzeiten.
- **Exact Official URLs:**
 - [SGB XII Full Statutory Text \(Gesetze im Internet\)](#)
 - [BMAS Grundsicherung im Alter Portal](#)
 - [BMAS Regelbedarfe Statistics](#)

Pillar 3: Macro-Demographics & Long-Term Aging (Destatis 16. BVB, 2024-2070)

- **Official Institution:** Federal Statistical Office of Germany (*Statistisches Bundesamt - Destatis*).
- **Official Dataset:** 16. koordinierte Bevölkerungsvorausberechnung (16. BVB, 2024-2070).
- **Exact Calibrated Parameters:**
- **2024 Population Baseline:** 84.67 million total (51.52M working-age 20-66; 17.50M elderly 67+; 6.20M aged 80+).
- **Old-Age Dependency Ratio (OADR):** Modeled across all 4 official variants from 34.0% in 2024 to:
 - Variant 1 (*Moderate G2-L2-W2*): 48.6% by 2070 (TFR 1.47, Net Migration +250k/yr).
 - Variant 2 (*Aging Shock G1-L3-W1*): 52.8% by 2070 (TFR 1.29, Net Migration +150k/yr, High Longevity).
 - Variant 3 (*High Migration G3-L2-W3*): 45.1% by 2070 (TFR 1.65, Net Migration +400k/yr).
 - Variant 4 (*Contraction G1-L1-W1*): 47.2% by 2070 (Low Fertility, Low Migration).
- **Life Expectancy at Age 67:** ~21.5 years for females vs. ~18.5 years for males (~20.0 years combined).
- **Exact Official URLs:**
 - [Destatis 16. BVB Official Overview](#)
 - [Destatis GENESIS-Online Database \(Table 12421\)](#)

Pillar 4: Household Wealth Distributions & Portfolios (Bundesbank PHF & DWA)

- **Official Institution:** Deutsche Bundesbank.
- **Official Datasets & Studies:**
 - Panel on Household Finances (PHF) Wave 2023 (Published April 2025).
 - Distributional Wealth Accounts (DWA 2024-2026).
- **Exact Calibrated Parameters:**
- **Net Wealth Medians (Mid-Career 35-44 / 45-54):**
 - German Natives: Median EUR 148,000-EUR 215,000 (Financial wealth baseline ~EUR 22,000).
 - General Migrants: Median EUR 42,000-EUR 68,000 (Financial wealth baseline ~EUR 5,300).
 - General Refugees: Median EUR 5,400-EUR 11,200 (Financial wealth baseline ~EUR 1,500).
 - Ukrainian War Refugees (2022+): Median EUR 4,500-EUR 7,800 (Financial wealth baseline ~EUR 420).

- *Ukrainian Migrants (Pre-2022)*: Median **EUR 42,000-EUR 74,000** (Financial wealth baseline ~EUR 4,900).
- **Asset Allocation Structures:**
 - *Liquid Bank Deposit Share*: 62%-65% (Natives) vs. 82%-91% (Refugees).
 - *Stock/ETF Participation Rate*: 38% (High-education Natives) vs. 15% (Refugees).
 - *Homeownership Rate*: 58% (Older Natives) vs. 2%-3% (Recent Refugees).
- **Exact Official URLs:**
 - [Bundesbank PHF Wealth Study \(April 2025\)](#)
 - [Bundesbank Distributional Wealth Accounts \(DWA\)](#)
 - [Bundesbank Research Data and Service Centre \(RDSC\)](#)

Pillar 5: Refugee Integration & Wage Dynamics (IAB-BAMF-SOEP)

- **Official Institutions:** *Institute for Employment Research (Institut für Arbeitsmarkt- und Berufsforschung - IAB Nürnberg) & Federal Office for Migration and Refugees (BAMF).*
- **Official Surveys & Reports:**
 - *IAB-BAMF-SOEP Survey of Refugees (2024-2026).*
 - *IAB-Forschungsbericht: Ukrainian Refugee Labor Integration Monitoring.*
- **Exact Calibrated Parameters:**
 - **Refugee Employment Integration Curve:** 18% (yr 1) $\rightarrow$ 46% (yr 3) $\rightarrow$ 63% (yr 5) $\rightarrow$ 73% (yr 8).
 - **Qualification Deskilling Disparity:** **-65.5% gross wage effect** for tertiary-educated refugees working in unaccredited, non-matching occupations.
 - **Language Progression (CEFR):** A1/A2 (25%), B1 (40%), B2/C (35%).
 - **Monthly Remittances to Country of Origin:** Ukrainian refugees (~EUR 180/mo female, ~EUR 120/mo male), General refugees (~EUR 140/mo), General migrants (~EUR 85/mo).
- **Exact Official URLs:**
 - [IAB-BAMF-SOEP Refugee Survey Portal](#)
 - [IAB Ukrainian Refugee Labor Monitoring](#)
 - [IAB Research Data Centre \(FDZ\)](#)

Pillar 6: Realized Historical Pension Outcomes & Gender Gaps (DRV Rentenatlas)

- **Official Institution:** *Deutsche Rentenversicherung Bund (DRV).*
- **Official Publication:** *Rentenatlas 2024/2025 & Statistikportal Rentenbestand / Rentenzugang.*
- **Exact Calibrated Parameters:**
 - **Contribution Years:** German Men (38.5 yrs), German Women (29.2 yrs), Migrant Men (25.1 yrs), Migrant Women (18.4 yrs).
 - **Realized EP Accumulation:** German Men (36.8 EP $\rightarrow$ EUR 1,378 net), German Women (24.5 EP $\rightarrow$ EUR 918 net).
 - **Social Safety Net Dependence Rates:** ~8.4% (German natives) vs. 28.5%-46.2% (Migrants/Refugees).
- **Exact Official URLs:**
 - [DRV Statistics & Rentenatlas Portal](#)

- [DRV Research Data Centre \(FDZ-RV\)](#)

Pillar 7: Macroeconomic Time Series & International Adequacy (Bundesbank, EU, OECD)

- **Official Institutions:** *Deutsche Bundesbank, European Commission (DG ECFIN), OECD.*
- **Official Series & Reports:**
 - *Bundesbank SDMX Macroeconomic Statistics (2000-2026).*
 - *European Commission 2024 Ageing Report.*
 - *OECD Pensions at a Glance 2023/2025.*
- **Exact Calibrated Parameters:**
 - **10-Year German Bund Yields:** Historical 2000-2026 series (2.15% in 2026).
 - **CPI Inflation Rates:** Historical 2000-2026 series (1.9%-2.0% in 2026 baseline).
 - **Capital Market Return Assumptions:** Real Equity Return = 4.5%-5.0% net of fees; Real Liquid Deposits = -0.5% net of inflation.
 - **Adequacy Target (B_{target}):** 60% baseline Net Replacement Rate (NRR) bounded between EUR 1,113 and EUR 1,800/month.
- **Exact Official URLs:**
 - [Bundesbank Time Series Database](#)
 - [European Commission 2024 Ageing Report](#)
 - [OECD Pensions at a Glance](#)